

Answer Key + Explanations

Q1 — Scale problem: why final maps miss small

objects

Correct: C

- **Why correct (2–3 sentences):** The assignment frames the scale problem as a tradeoff: deeper CNN layers gain semantic richness but lose spatial resolution due to pooling/strides. Small objects occupy too few cells on the final map and become poorly represented for localization (and sometimes effectively invisible).
 - **Why A is a plausible pitfall:** It sounds reasonable if you associate “small” with “hard to classify,” but the assignment explicitly says later layers have **richer** semantics; the main issue is **resolution**, not semantics.
 - **Why B is a plausible pitfall:** “Not enough context” is a common detection failure mode, but deeper layers generally have **larger** receptive fields; the problem is that small objects are *subsumed* by the receptive field and pooled away.
 - **Why D is a plausible pitfall:** NMS can suppress close detections in dense scenes, but the assignment identifies the stubborn core issue as the semantic–resolution mismatch in backbone features, not post-processing.
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Q2 — Lateral 1×1 convs: why they’re necessary

Correct: B

- **Why correct:** The assignment describes lateral connections as using **1×1 convolutions** to project bottom-up features into a **shared channel dimension**

(e.g., 256) so that **element-wise addition** with the top-down features is possible. Conceptually, this also selects a “useful subspace” of each backbone level before fusion.

- **Why A is a plausible pitfall:** People often conflate “matching shapes” with “upsampling,” but 1×1 convolution changes **channels**, not **spatial resolution**.
 - **Why C is a plausible pitfall:** FPN feels like it “fixes” downsampling, but it doesn’t remove pooling/strides; it compensates by injecting semantics into finer maps.
 - **Why D is a plausible pitfall:** Receptive field growth is determined by backbone depth/strides. Lateral 1×1 layers don’t control receptive field expansion; they align and transform features for fusion.
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Q3 — Top-down pathway: what it contributes

Correct: A

- **Why correct:** The assignment explains the top-down pathway as a second pass that **upsamples coarser, semantically rich maps** and propagates that semantic context to higher-resolution levels. This directly targets the semantic weakness of shallow maps while preserving their spatial detail.
 - **Why B is a plausible pitfall:** Attaching multiple detection heads can be part of multi-scale detection, but it doesn’t explain **how** semantics are transferred across scales; FPN’s key is **top-down + lateral fusion**.
 - **Why C is a plausible pitfall:** Historically correct approach (featurized image pyramids), but the assignment highlights its compute cost and positions FPN as the efficient alternative.
 - **Why D is a plausible pitfall:** Upsampling artifacts and aliasing are real issues (and motivate later extensions), but the top-down pathway’s primary purpose in the assignment is *semantic propagation*, not specifically large-object performance.
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Q4 — Worked example dimensions

Correct: C ($4\times 4\times 256$)

- **Why correct:** The 1×1 conv projects C4 from 1024 channels to 256 while keeping spatial size at 4×4 ; the following 3×3 conv preserves spatial size and channel count, yielding $P4 = 4\times 4\times 256$.
 - **Why A is a plausible pitfall:** It ignores the explicit channel projection to the pyramid's target dimension.
 - **Why B is a plausible pitfall:** It applies the upsampling step too early; upsampling is used when producing the next finer level (e.g., contributing to P3), not to define P4.
 - **Why D is a plausible pitfall:** It mixes up backbone outputs (C3) with pyramid outputs (P4), a common confusion when first learning FPN notation.
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Q5 — What element-wise addition assumes

Correct: A

- **Why correct:** Addition requires strict spatial alignment and equal channel dimensionality; this implies the model treats corresponding positions/channels as compatible representations. The assignment's discussion of lateral 1×1 projection supports this: it makes the tensors comparable so they can be summed.
 - **Why B is a plausible pitfall:** It's tempting to assume "fusion" means global interaction, but the assignment (and LFPN motivation) highlights that element-wise fusion is **local/point-wise**, not long-range.
 - **Why C is a plausible pitfall:** Learners may interpret the higher-resolution map as "more trustworthy," but addition blends signals rather than overriding one with the other.
 - **Why D is a plausible pitfall:** FPN being backbone-independent is an empirical/architectural observation; addition itself doesn't guarantee it.
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Q6 — Level assignment for small objects

Correct: B

- **Why correct:** The example shows very small objects can yield k below the pyramid range, then get clipped to the **finest** level (P2). This reflects the design: small objects need high spatial resolution for localization, while semantics are recovered via top-down injection.
 - **Why A is a plausible pitfall:** It recognizes “semantics matter,” but contradicts the assignment’s key point that coarse maps lack spatial detail and thus are poor for small-object localization.
 - **Why C is a plausible pitfall:** Image pyramids are a known multi-scale technique, but the assignment explicitly contrasts them with FPN as too expensive.
 - **Why D is a plausible pitfall:** Random assignment can be used in some training tricks, but the assignment’s intent is deterministic size-to-level routing.
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Q7 — Efficiency vs. image pyramids

Correct: B

- **Why correct:** The assignment notes featurized image pyramids are expensive because compute grows quickly with scale, while FPN builds multi-scale representations **from features the backbone already computed**, adding comparatively small overhead.
 - **Why A is a plausible pitfall:** Parallelism doesn’t remove the fundamental cost of multiple backbone passes; the assignment’s critique is about total compute.
 - **Why C is a plausible pitfall:** Some detectors avoid early layers due to weak semantics, but FPN relies on shallow maps (strengthened via fusion) to recover localization.
 - **Why D is a plausible pitfall:** FPN uses 1×1 and 3×3 layers in the neck; the backbone dominates compute and 3×3 smoothing remains part of the design.
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Q8 — GraphFPN: what it changes

Correct: A

- **Why correct:** The assignment describes GraphFPN as challenging the **fixed grid topology** assumption: objects cluster and overlap in ways a static grid interaction pattern can't reflect. GraphFPN uses a superpixel hierarchy and graph edges (contextual + hierarchical) to enable richer within- and across-scale interactions.
 - **Why B is a plausible pitfall:** "Multi-scale = image pyramid" is a common conflation, but GraphFPN focuses on feature interaction topology, not rerunning the backbone.
 - **Why C is a plausible pitfall:** This echoes the original semantic weakness of shallow layers, but GraphFPN's stated critique is about *interaction structure*, not semantics per se.
 - **Why D is a plausible pitfall:** Anchor assignment is a real multi-scale mechanism, but it's not GraphFPN's core contribution in the assignment summary.
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Q9 — LFPN: why latent space + GCN

Correct: B

- **Why correct:** LFPN is motivated by the assignment's point that element-wise fusion (addition/concat/weighted multiplication) cannot model **long-range dependencies**—distant but semantically related pixels don't interact. LFPN projects features to a low-dimensional latent space, applies GCN-based fusion (ISF/CSF), then projects back.
 - **Why A is a plausible pitfall:** Upsampling-stage semantic weakening is a real issue (highlighted under MSRA-FPN), so it's believable—but it's not LFPN's stated target.
 - **Why C is a plausible pitfall:** Oriented boxes are important in remote sensing, but LFPN's motivation is dependency modeling, not geometry conventions.
 - **Why D is a plausible pitfall:** RoI Align addresses quantization errors in region pooling (noted in the remote sensing discussion), but LFPN is a pyramid fusion redesign.
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Q10 — DM-FPN: why remote sensing is different

Correct: B

- **Why correct:** The assignment emphasizes VHR images can be thousands of pixels wide, include extreme scale imbalance (tiny vehicles vs. huge structures), lack canonical orientations, and use oriented bounding boxes—conditions that stress standard pipelines and motivate DM-FPN-style adaptations.
- **Why A is a plausible pitfall:** It describes a “clean” detection regime and misses the assignment’s stated sources of difficulty.
- **Why C is a plausible pitfall:** It repeats the generic scale problem framing but ignores the *domain constraints* (tiling, density, orientation, box type) that the assignment highlights.
- **Why D is a plausible pitfall:** Parameter tweaks to assignment formulas can matter, but the assignment presents DM-FPN as addressing a broader stack (e.g., training/inference tiling, NMS adaptation, pooling quantization).